

AYA AZ YASIN

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Research Interests

Liquid–vapor phase change and interfacial transport; thin-film and contact-line dynamics; multiscale modeling coupling molecular dynamics, continuum CFD, and kinetic theory; with applications in cryogenic propellant management, microgravity thermal systems, and heat-transfer technologies.

Education

University of Cincinnati, Cincinnati, OH PhD in Mechanical Engineering Advanced to candidacy (Jun 2026)	2024 – present
University of Cincinnati, Cincinnati, OH MS in Aerospace Engineering	2024
University of Cincinnati, Cincinnati, OH BS in Mechanical Engineering Technology Minor in Mathematics	2022

Peer-Reviewed Journal Publications

4. A. Sarchami, S. Pakanati, **A. Yasin**, M. Jog, and K. Bellur, (2026) “An experimental study of interfacial dynamics control using temperature-sensitive surfactants”, *Langmuir*, 42 (20), doi:10.1021/acs.langmuir.5c06748
3. **A. Yasin***, S. Pakanati*, and K. Bellur, (2026) “A multiscale CFD model of evaporating hydrogen menisci: incorporating subgrid thin-film dynamics and in situ accommodation coefficients”, *Fuels*, 7(1), doi:10.3390/fuels7010003
2. **A. Yasin** and K. Bellur, (2025) “Computational modeling of evaporation without tuning coefficients”, *Applied Thermal Engineering*, 276, doi:10.1016/j.applthermaleng.2025.126807
1. U. Chakrabarti*, **A. Yasin***, K. Bellur, and J. Allen, (2023) “An investigation of phase change induced Marangoni-dominated flow patterns using the Constrained Vapor Bubble data from ISS experiments”, *Frontiers in Space Technologies*, 4, doi:10.3389/frspt.2023.1263496

*equal contribution authors

Manuscripts in Preparation

1. **A. Yasin**, T. Collins, and K. Bellur, “Probing the origins of the inequality of accommodation coefficients in non-equilibrium liquid-vapor phase change”.

Grants and Fellowships

University Research Council Graduate Student Stipend Award (\$8,500) UC Office of Research Competitively awarded to fund a student-faculty collaborative research proposal.	2026
Lester Nenninger Award (\$1,000) UC College of Engineering and Applied Science Awarded for research in computational science and engineering.	2026

Research Fellowship (\$1,000) UC Graduate Student Government	2026
Prof. Kirti Ghia Endowed Graduate Student Scholarship (\$1,000) UC Dept of Mechanical, Materials, and Industrial Engineering, Awarded for research in computational fluids.	2025
Undergraduate Research Fellowship (\$750) UC Office of Research	2022

Honors and Awards

Graduate Engineer of the Month, UC College of Engineering and Applied Science	Jul 2027
Excellence in Teaching Award (Honorable Mention), UC Graduate College	2024
Travel Award, APS, Division of Fluid Dynamics	2023
Outstanding Senior Award, UC College of Engineering and Applied Science	2022

Invited Talks and Panels

3. Invited Panelist, Graduate Student Teaching Orientation Aug 2026
Center for the Enhancement of Teaching and Learning (CET&L), University of Cincinnati
2. Student Panelist, Departmental Graduate Student Orientation 2025, 2026
Dept of Mechanical, Materials, and Industrial Engineering, University of Cincinnati
1. Liquid–vapor phase change in aerospace applications Apr 2025
Invited talk at graduate seminar, Dept of Aerospace Engineering, University of Cincinnati

Conference Presentations

11. **A. Yasin**, E. Stewart, and K. Bellur, “Breakdown of lubrication theory in evaporating thin films: inertial and phase-change effects”, 79th APS Division of Fluid Dynamics Annual Meeting, Orlando, FL, Nov 2026
10. **A. Yasin**, U. Chakrabarti, and K. Bellur, “Modeling multiscale oscillations in thin liquid films”, 11th ASTFE Thermal and Fluids Engineering Conference, Tempe, AZ, Mar 2026
9. A. Sarchami, S. Pakanati, T. Enam, **A. Yasin**, and K. Bellur, “Stability and contact line dynamics of evaporating thin liquid films”, 11th ASTFE Thermal and Fluids Engineering Conference, Tempe, AZ, Mar 2026
8. **A. Yasin**, U. Chakrabarti, and K. Bellur. “Multiscale oscillations in thin liquid films”, (poster) ASME International Mechanical Engineering Congress & Exposition, Memphis, TN, Nov 2025
7. **A. Yasin** and K. Bellur, “Exploring two-dimensional flows in evaporating thin films: a step towards a dynamic model” 10th ASTFE Thermal and Fluids Engineering Conference, Washington, DC, Mar 2025
6. **A. Yasin** and K. Bellur, “Modeling of evaporation in cryogenic fuels without tuning coefficients”, 35th NASA Thermal and Fluids Analysis Workshop, Cleveland, OH, Aug 2024
5. **A. Yasin** and K. Bellur, “Modeling evaporation without tuning coefficients”, 51st Midwestern University Fluid Mechanics Retreat, Rochester, IN, Apr 2024
4. **A. Yasin** and K. Bellur, “A numerical study of coefficient-free kinetic evaporation modeling in liquid hydrogen”, 76th APS Division of Fluid Dynamics Annual Meeting, Washington, DC, Nov 2023
3. **A. Yasin**, U. Chakrabarti, K. Bellur, and J. Allen, “An investigation of Marangoni-induced flow in Constrained Vapor Bubble ISS experiments”, 50th Midwestern University Fluid Mechanics Retreat, Rochester, IN, Apr 2023

2. **A. Yasin** and K. Bellur, “A CFD model of evaporation in liquid hydrogen without the need for tuning coefficients”, 75th APS Division of Fluid Dynamics Annual Meeting, Indianapolis, IN, Nov 2022
1. **A. Yasin**, R. Gilligan, D. Heitmeyer, and K. Cohen, “A solution to the 2022 AUVSI Student Unmanned Aerial Systems competition”, AIAA Region III Student Conference, Purdue University, West Lafayette, IN, Mar 2022

underline shows presenter

Research Experience

Research Assistant, Lab for Interfacial Dynamics (advised by Dr. Kishan Bellur) 2022 – present
 Dept of Mechanical, Materials, and Industrial Engineering, University of Cincinnati, Cincinnati, OH

- Developed coefficient-free models of liquid–vapor phase change, replacing empirically tuned accommodation coefficients with theoretical predictions in CFD (Ansys) implementations.
- Applied these models to cryogenic propellant management and microgravity thermal systems, validating against ISS Constrained Vapor Bubble data.
- Analyzed interferometry and optical-thermometry data from evaporating and oscillating thin-film experiments, and interfacial-curvature-switching experiments using temperature-sensitive surfactants.

Research Assistant, Procter & Gamble Digital Accelerator Fall 2022
 University of Cincinnati, Cincinnati, OH

- Built genetic-algorithm methods for computing arbitrarily oriented bounding boxes for image-processing in micro-CT applications.

Industry Experience

Product Development Engineering Co-op Spring & Summer 2021
 GMi Companies, Lebanon, OH

Manufacturing Engineering Co-op Spring & Fall 2019
 Regal Rexnord, Florence, KY

Teaching Experience

Instructor of record

Applied Computational Methods – Lecture & Lab (undergraduate) Spring 2025
 Foundations of Engineering Design Thinking II (undergraduate) Spring 2024
 Foundations of Engineering Design Thinking I (undergraduate) Spring 2023, Fall 2023

Teaching assistant

Thermal Environmental Systems & Heat Transfer Lab (undergraduate) Spring 2026
 Foundations of Engineering Design Thinking II (undergraduate) Spring 2022
 Foundations of Engineering Design Thinking I (undergraduate) Fall 2020, Fall 2021

Guest lectures

Computational Fluid Dynamics (UDFs in Ansys Fluent) (graduate) Spring 2024, Spring 2026
 Thermodynamics (Entropy) (undergraduate) Fall 2024

Mentoring and Supervision

Mentored undergraduate research co-ops and REU students. 2024 – present
 Supervised a team of six undergraduate and two graduate teaching assistants. 2023 – 2024
 Mentored undergraduate students in the First-Year Engineering program. 2023 – 2024
 Served as a peer mentor at the APS DFD meeting. 2026

Service

Served on a university-wide committee to develop standards for graduate handbooks. 2026
Mechanical Engineering Chair, Department Graduate Student Association. 2026 – present
Organized a graduate seminar series featuring student research talks and faculty-led professional-development sessions. 2026 – present

Reviewer for: Physics of Fluids; ACS Nano Letters; ASME Journal of Heat and Mass Transfer; AIAA Journal of Thermophysics and Heat Transfer; Communications in Theoretical Physics.

Outreach

T. Collins, **A. Yasin**, K. Bellur, and E. Payton, “Bubbles and Biases: Demystifying Research and Sustainability”, doi:10.7945/y6m5-2f09, Jul 2026
Public-audience podcast on phase-change research, created by a mentored REU student.

Skills

Programming: MATLAB, Python, C, C++, Bash, VBA, Git, $\text{\LaTeX}$.
Simulation: Ansys Fluent, OpenFOAM, STAR-CCM+; COMSOL, FEniCSx; LAMMPS.
HPC & Tools: MPI, Slurm, LabVIEW, SolidWorks, Simcenter 3D.